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Quantum-inspired post-quantum secure DCT watermarking for tamper detection and localization in medical images

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ABSTRACT

As medical imaging becomes increasingly central to telemedicine and electronic health records; robust integrity verification is required to preserve diagnostic quality while enabling precise tamper detection. Existing quantum image watermarking methods lack post-quantum key security and deliberate fragility. This paper proposes a quantum-inspired watermarking framework for medical images with post-quantum key security and deliberate fragility based on entangled DCT coefficients. Watermarks are carried by collective coefficient sums, making any alteration detectable. Post-quantum security is ensured through SHA3-256 key derivation. Experiments on three medical datasets (ChestX-ray14, BraTS, OCTID) show excellent imperceptibility (PSNR 44.2 dB, SSIM >0.98), high tamper sensitivity (BER >30%), and precise localization (accuracy >80%). Unlike prior methods, the framework integrates post-quantum security, deliberate fragility, and tamper localization. Furthermore, a lightweight Random Forest classifier is trained on global and block-wise BER statistics to automatically distinguish intact images from those subjected to benign processing or malicious tampering, enhancing the system's practical utility as an AI-assisted integrity verification tool for medical images in future quantum-threat environments.

1. Introduction

The expanding role of digital medical imaging in telemedicine, cloud-based diagnostics, and electronic health records has brought issues of data authenticity, integrity, and protection to the forefront of medical information security [1]. Medical images such as MRI, CT scans, and X-rays contain highly sensitive diagnostic information that must remain unaltered throughout transmission across potentially insecure networks [2]. Unauthorized modifications, whether malicious or accidental, can lead to misdiagnosis with potentially severe clinical consequences. Consequently, robust mechanisms are required to embed verification data directly into the image itself without compromising its diagnostic value [3]. Digital image watermarking has emerged as a prominent solution for these security requirements, enabling the imperceptible embedding of metadata, patient identifiers, or cryptographic signatures within the host image [4]. Among existing approaches, spatial-domain and block-based watermarking techniques have been widely adopted due to their simplicity, high embedding capacity, and compatibility with medical imaging workflows [5]. However, conventional watermarking approaches often rely on fixed embedding patterns or heuristic parameter selection, making them vulnerable to advanced attacks and limiting their ability to balance imperceptibility and robustness [6]. Recent studies have highlighted that medical images exhibit heterogeneous regions, where uniform embedding strategies may introduce unnecessary distortion or insufficient robustness depending on local characteristics [7]. Adaptive watermarking strategies have been proposed to address this issue by dynamically selecting embedding parameters based on image content,

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improving both visual quality and security performance [8]. Nevertheless, many existing methods primarily focus on imperceptibility or robustness independently, without jointly addressing tamper detection and localization capabilities [9]. Forgery detection represents another critical challenge, particularly for localized tampering attacks in medical images. Block-based tamper detection combined with correlation or similarity analysis has demonstrated promising performance in identifying altered regions with high accuracy [10]. In addition, several schemes have explored the integration of recovery information to restore tampered regions, but these approaches often suffer from limited robustness or dependency on embedded recovery data [11]. Moreover, hybrid transform-domain watermarking methods combining DCT, wavelet, or SVD techniques have shown improved robustness and imperceptibility, but they typically increase computational complexity and lack precise localization mechanisms [12]. Despite these advances, most existing methods do not account for the long-term security threat posed by future quantum computers, which could compromise classical cryptographic primitives currently used for key management and authentication. Recent research in quantum image watermarking has explored alternative paradigms leveraging quantum mechanical properties such as superposition, entanglement, and parallelism. However, existing approaches still lack integration of post-quantum cryptographic security and deliberate fragility for accurate tamper localization. This motivates the development of a unified framework that combines post-quantum security, adaptive embedding principles, and precise tamper detection mechanisms. The proposed approach introduces three key innovations. First, a post-quantum secure key generation mechanism ensures long-term protection against quantum adversaries, binding the watermarking parameters to the image context through cryptographic hashing. Second, a quantum-inspired coefficient entanglement mechanism embeds the watermark through differential modulation of coefficient groups, creating structural dependencies that make any alteration no matter how subtle detectable. This deliberate fragility transforms the watermark into a sensitive probe of image integrity, mirroring the quantum observer effect where interaction inevitably perturbs the state. Third, the block-wise DCT embedding enables precise tamper localization, providing forensic analysts with a map of altered regions without compromising imperceptibility. This paper makes the following contributions.

- A first medical image watermarking scheme is proposed that combines post-quantum secure key generation (SHA3-256) with deliberate fragility.
- Tampering is made detectable without error-correcting codes by means of quantum-inspired coefficient entanglement.
- Precise tamper localization is achieved while imperceptibility is preserved.
- A lightweight machine learning decision layer is integrated to classify images into three categories using global and block-wise BER statistics as features, thereby bridging the gap between signal-level tamper detection and practical forensic decision-making.
- A formal security analysis against Grover's algorithm is provided, along with compliance with NIST standards.

The remainder of this paper is organized as follows. Section 2 reviews related work in quantum image watermarking, categorizing existing approaches and highlighting their respective strengths and limitations. Section 3 details the proposed methodology, including post-quantum key generation, quantum-inspired coefficient entanglement embedding, and the extraction and tamper detection mechanism. Section 4 presents experimental results evaluating imperceptibility, robustness, and tamper localization capabilities. Finally, Section 5 concludes the paper and discusses directions for future research.

2. Related work

Quantum image watermarking has emerged as a promising research direction at the intersection of quantum information processing and multimedia security [13]. Unlike classical watermarking techniques, quantum watermarking leverages quantum mechanical properties such as superposition, entanglement, and parallelism to achieve enhanced security, processing speed, and imperceptibility [14]. This section reviews representative quantum image watermarking approaches, organized by their core strategies. Geometric transformation-based methods have been explored by Iliyasu et al. [15], who proposed WaQI, a keyless and blind watermarking strategy based on restricted geometric transformations (rGTQI) within the FRQI framework. Their scheme preserves watermarked image quality by confining transformations to sub-areas that withstand distortion, and the reversible nature of quantum circuits enables original image recovery without a key. However, the approach assumes fault-tolerant quantum hardware and requires classical image pre-processing. Cui et al. [16] extended geometric transformations by combining space geometric transformation with Fibonacci scrambling to improve robustness against noise and cropping attacks, embedding a scrambled watermark into the two least significant bits using NEQR representation. This method demonstrates superior robustness against large-scale cropping but achieves lower imperceptibility and involves complex quantum circuit design. Quantum transform and chaotic mapping methods have been proposed by Yuan et al. [17], who integrated quantum discrete wavelet transform (QDWT) with Sinusoidal-Tent mapping and Baker mapping. Their algorithm uses a chaotic sequence to determine the embedding channel and embeds the scrambled watermark into the diagonal (HH) subband, achieving significantly lower computational complexity than classical DWT. However, it requires precise synchronization of initial parameters. Jiang et al. [18] proposed a scheme combining quantum Hilbert scrambling with Moiré fringe steganography, achieving good visual quality with lower time complexity but primarily focusing on grayscale images. Quantum representation and encoding methods include the work of Aly [19], who proposed a quantum color image watermarking technique (NQCIWT) based on the New Quantum Binary Image Detector (NQBID) representation using Bloch sphere encoding. The approach offers high imperceptibility and resistance to hacking attacks but relies on key storage with limited validation on larger images. Dong et al. [20] proposed a scheme based on quantum error correction codes (QECC) to address pixel loss and image distortion, using three-qubit encoding to provide robustness against phase-flip errors, though it is limited by fixed image dimensions. Noise masking and sparse approximation methods have been investigated by Pan et al. [21], who proposed a reversible watermarking scheme for radiographic images that masks the watermark into quantum noise rather than modifying diagnostically significant regions. The scheme achieves exceptional imperceptibility with minimal distortion but is fragile by design, limiting robustness against intentional attacks. Agrawal et al. [22] proposed a secure image authentication technique combining Sparse Approximation (SA) with Quantum Encryption (QE), achieving high embedding capacity and robust performance under various attacks, but requiring non-singular measurement matrices and lacking validation on quantum hardware. Table 1 summarizes the advantages and limitations of these representative methods. Despite significant progress, existing quantum image watermarking methods face several common challenges. Geometric transformation-based approaches either assume fault-tolerant quantum hardware or achieves lower imperceptibility. Quantum transform and chaotic mapping methods suffer from sensitivity to initial parameter selection. Quantum representation-based techniques rely on key storage with limited validation. Quantum error correction-based schemes are constrained by fixed image dimensions. A common limitation across all these approaches is the absence of a systematic mechanism to achieve deliberate fragility with precise tamper localization. Furthermore, none of these methods integrate post-quantum cryptographic primitives for

Table 1
Comparison of quantum image watermarking methods.

Author(s)	Method	Advantages	Limitations
Iliyasu et al. [15]	Restricted geometric transformations (rGTQI) + FRQI	Blind, keyless, reversible authentication	Requires fault-tolerant quantum hardware, classical pre-processing needed
Pan et al. [21]	Quantum noise masking + histogram shifting (spatial + wavelet)	Reversible, preserves diagnosis value, minimal distortion	Fragile (not robust to attacks), limited capacity, medical image specific
Agrawal et al. [22]	Sparse approximation + quantum logistic map + DWT-SVD	High capacity, robust to multiple attacks, strong encryption via quantum chaos	Requires non-singular matrices, not validated on quantum hardware
Aly [19]	NQBID (block sphere) + polar form + random scrambling	High imperceptibility, resistant to hacking attacks, supports multiple color spaces	Requires key storage, limited validation on larger images
Cui et al. [16]	GTA (QIS + QIR) + Fibonacci scrambling + QMF	Robust to large-scale cropping, good noise resistance, low complexity	Complex quantum circuit design
Yuan et al. [17]	QDWT + ST mapping + baker mapping + HH subband embedding	Large key space, low computational complexity, strong security	Sensitive to initial parameter selection, requires chaotic sequence synchronization
Dong et al. [20]	QECC (3-qubit) + NEQR + LSB + Arnold transform	Robust to phase-flip errors, high visual quality, low BER	Fixed image sizes, limited to binary watermarks
Jiang et al. [18]	Quantum Hilbert scrambling + Moiré fringe steganography	High security, lower time complexity, good visual quality	Grayscale image focus, complex circuit design

key management, leaving the overall system vulnerable to future quantum adversaries. This paper addresses these gaps by proposing a framework that combines post-quantum secure key management with quantum-inspired coefficient entanglement, where the watermark is carried by collective properties of coefficient groups, making any alteration detectable while maintaining high visual quality.

3. Proposed approach

This section details the proposed watermarking scheme, which combines three essential characteristics: invisibility through frequency-domain embedding, post-quantum cryptographic security for key management, and a quantum-inspired fragility that enables precise tamper detection. The system is divided into two main processes embedding and extraction each designed to be the inverse of the other. The exposition follows the logical flow from key generation to the final decision, with careful justification of each design choice. The architecture of the scheme is illustrated in Fig. 1. The overall design of the proposed framework is consistent with recent watermarking systems that adopt block-based processing, key-driven embedding, and structured tamper detection strategies [23]. In particular, the use of pseudo-random selection mechanisms and hash-based key generation enhances resistance against unauthorized access and improves unpredictability of embedding locations. Furthermore, block-wise processing enables fine-grained tamper localization, which has been shown to significantly improve detection accuracy in medical image authentication scenarios [24]. The embedding stage begins with the generation of a secret key K using a post-quantum secure pseudo-random number generator (CSPRNG). This key uniquely determines both the locations where the watermark will be inserted in the Discrete Cosine Transform (DCT) domain and the watermark symbols themselves. The host image is first transformed block-wise with the DCT, and the watermark is embedded by additively modifying selected mid-frequency coefficients. The inverse DCT then reconstructs the watermarked image, which is indistinguishable from the original to the human eye. The extraction and tamper-detection stage is the inverse of the embedding process but introduces a quantum-inspired principle: the watermark is made deliberately fragile so that any alteration to the image or any attempt to extract the watermark without the correct key produces a detectable incoherence. An authorized user, possessing the key K, can regenerate the exact embedding parameters and, using a stored reference of the pristine watermarked image, extract the watermark from a test image. The Bit Error Rate (BER) between the extracted watermark and the original serves as a sensitive tamper indicator. This approach mirrors the quantum measurement postulate, where the act of observation perturbs the state; here, any "interaction" (tampering) irreversibly alters the watermark, while extraction with the correct key leaves it unchanged.

3.1. Post-quantum secure key generation and watermark definition

The security of the entire scheme rests on the secrecy of the key K. To guarantee long-term protection against adversaries equipped with quantum computers, we adopt a post-quantum secure key generation mechanism. The master key K is derived from a secret seed K_s and optional image-specific metadata (such as dimensions or a checksum) using a cryptographic hash function that is believed to be quantum-resistant. Formally:

$$K = \text{SHA3-256}(K_s \parallel I_{\text{meta}}) \tag{1}$$

Where SHA3-256 is the NIST-standardized hash function, and I_{meta} binds the key to a particular image context, preventing cross-image attacks.

The entropy source for K_s is a hardware random number generator, ensuring uniform distribution. A 256-bit key provides a security level of 128 bits against Grover's algorithm, which is widely accepted as sufficient for post-quantum cryptography [25]. To transform K into a reproducible embedding strategy, we use a CSPRNG seeded with K. The CSPRNG is instantiated with a stream cipher (AES-256 in CTR mode) that is believed to remain secure against quantum attacks when used

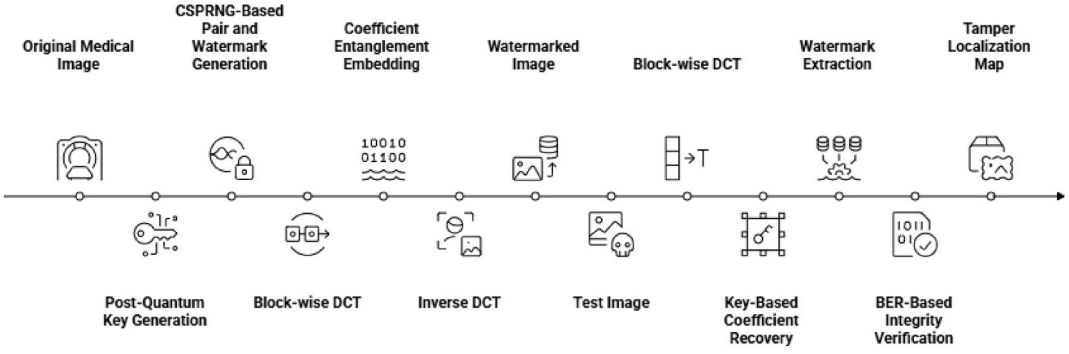


Fig. 1. Image watermarking and integrity verification process.

with a key to sufficient length. From this CSPRNG we generate two independent sequences. First, we generate a set of N pseudo-random coordinate pairs which indicate the positions of the DCT coefficients to be modified:

$$\{(u_i, v_i)\}_{i=1}^{N}, u_i, v_i \in \{0, 1, \ldots, 7\} \tag{2}$$

To balance invisibility and robustness, we restrict the selection to the mid-frequency range by rejecting any pair where $(u_i, v_i) = (0, 0)$ (the DC component) or where the sum $u_i + v_i$ is less than 2 or greater than 10. This heuristic is based on the well-known property of the DCT: modifications in very low frequencies cause visible blocking artifacts, while modifications in very high frequencies are easily removed by lossy compression. By focusing on mid frequencies, we achieve a watermark that is both perceptually invisible and resilient to standard image processing. Second, the same CSPRNG produces a bipolar watermark sequence:

$$W = \{w_i\}_{i=1}^{N}, w_i \in \{-1, +1\} \tag{3}$$

Where each w_i is obtained by taking the most significant bit of the CSPRNG output and mapping it to $+1$ for 1 and -1 for 0.

The use of a single key to generate both the locations and the values creates a cryptographically strong binding: without the key, an adversary cannot determine where the watermark is hidden nor what the expected symbols are. The framework can be extended with lattice-based signatures to authenticate the watermark origin, but this is not mandatory for the core tamper-detection functionality.

3.2. Quantum-inspired watermark embedding via coefficient entanglement

The originality of our approach lies in how the watermark is inserted. Instead of independent additive modulation of each coefficient, we introduce a mathematically controlled entanglement mechanism between coefficients, directly inspired by the quantum measurement postulate. The goal is to make the watermark inherently fragile: any subsequent modification of the image even a slight one breaks the correlations imposed during embedding, producing a detectable inconsistency during extraction. In this way, fragility is not merely an extraction-side property but a structural feature of the watermark itself.

3.2.1. DCT domain selection and coefficient pairing

We work in the DCT domain with 8×8 blocks, for the classical reasons of energy compaction and JPEG compatibility [26]. The image is first divided into non-overlapping blocks of size 8×8. For each block B with pixel values $b[x, y]$ $(x, y = 0, \ldots, 7)$, the two-dimensional DCT coefficients are given by Ref. [27]:

$$B(u, v) = c(u)c(v) \sum_{x=0}^{7} \sum_{y=0}^{7} b[x, y] \cos\left(\frac{(2x+1)u\pi}{16}\right) \cos\left(\frac{(2y+1)v\pi}{16}\right) \tag{4}$$

Where:

$$c(k) = \begin{cases} \frac{1}{\sqrt{8}}, & k = 0 \\ \frac{1}{2}, & k > 0 \end{cases} \tag{5}$$

This transform decorrelates the image data and concentrates energy into low-frequency coefficients, enabling us to embed information in perceptually less significant components [28]. For each block, we select a set of coefficient pairs (p, q) in the mid-frequency range (excluding the DC component and very high frequencies) using the CSPRNG seeded with the key K. The selection is deterministic yet unknown without the key. The innovation consists of not treating each coefficient independently. Instead, we form entangled coefficient pairs: for each watermark bit $w_i \in \{-1, +1\}$, we choose two coefficients $B(u_1, v_1)$ and $B(u_2, v_2)$ that will be modified jointly. The pair is generated by the CSPRNG so that the two positions are distinct but linked by the same random seed.

3.2.2. Embedding via differential modulation with correlation constraint

We first formally define the notion of "quantum-inspired" as used in this work. Unlike quantum-implemented watermarking methods that rely on quantum hardware (qubits, superposition, gates), a quantum-inspired approach is a classical algorithm that mimics structural properties of quantum systems using deterministic mathematical constructs.

Definition 1. (Coefficient entanglement): Let C = {C1,C2, …,Cn}C = {$C1$,$C2$, …,Cn} be a set of DCT coefficients. A subset $G \subseteq$C is said to be entangled if there exists a constraint function f:R|G|→R such that f(G) is constant and the watermark information is encoded in the controlled perturbation of this constraint while preserving orthogonal components. In this work, for pairs (n= 2), the constraint function is defined as the sum f($C1$,$C2$)=$C1$+$C2$, and the orthogonal component is the difference g ($C1$,$C2$)=$C1$−$C2$. The watermark is embedded by modifying f while keeping g invariant. This formulation distinguishes our approach from heuristic coefficient pairing: the dependency between coefficients is explicitly defined as a constraint subspace, analogous to correlated quantum states. The goal is to make the watermark inherently fragile: any subsequent modification to the image that perturbs the constraint f (even slightly) breaks the entanglement, producing a detectable inconsistency during extraction. Fragility thus becomes a structural feature, not just an extraction-side property. Using the entanglement formalism from Definition 1, for each watermark bit w_i we select a pair of coefficients (C_a,C_b) that will be treated as an entangled group. Let (C_a, C_b) be a pair of coefficients corresponding to a single bit w_i. We define the sum $S = C_a + C_b$ and the difference $D = C_a - C_b$. Embedding is performed by imposing a target relationship on the sum while keeping the difference unchanged (or vice-versa), thereby creating a dependency between the two coefficients. Concretely, we choose a target sum S_{target} based on the original sum S_{orig} and the bit w_i:

$$S_{\text{target}} = S_{\text{orig}} + \beta \cdot w_i \tag{6}$$

Where β is an embedding strength factor (distinct from the α used in independent additive schemes).

We then adjust the two coefficients to achieve this sum while preserving their original difference $D_{\text{orig}} = C_a - C_b$. The new values are:

$$C_a' = \frac{S_{\text{target}} + D_{\text{orig}}}{2}, C_b' = \frac{S_{\text{target}} - D_{\text{orig}}}{2} \tag{7}$$

This operation guarantees that the sum is modulated by the watermark bit, while the difference remains intact. Thus, the watermark is carried by a collective property (the sum) rather than by an individual coefficient. For each watermark bit, the embedding therefore proceeds as follows. Using the CSPRNG initialized with the secret key K, a pair of positions (u_1, v_1) and (u_2, v_2) is generated within the same block (or across two blocks, depending on a predefined mode). The original coefficients $C_1 = B(u_1, v_1)$ and $C_2 = B(u_2, v_2)$ are retrieved, and their sum $S_{\text{orig}} = C_1 + C_2$ and difference $D_{\text{orig}} = C_1 - C_2$ are computed. The target sum is then set to $S_{\text{target}} = S_{\text{orig}} + \beta \cdot w_i$, and the two coefficients are updated to $C_1' = (S_{\text{target}} + D_{\text{orig}})/2$ and $C_2' = (S_{\text{target}} - D_{\text{orig}})/2$. This process is repeated for all N watermark bits. The complete embedding procedure is summarized in Algorithm 1.

Algorithm 1 – Quantum-inspired watermark embedding procedure

Input: Host image I (size M × N), secret seed K_s, embedding strength β, watermark length N
Output: Watermarked image I'_wm
1. K ← SHA3-256(K_s ‖ I_meta)
2. Initialize CSPRNG with seed K
3. Divide I into 8 × 8 blocks B_k

(continued on next page)

(*continued*)

4. For each block B_k: a. Compute DCT: B_k(u,v) using Eq. (4)
5. For i = 1 to N:
a. Generate random block index k, pair positions (u_1,v_1), (u_2,v_2) within mid-frequency range
b. Retrieve coefficients C_1 = B_k(u_1,v_1), C_2 = B_k(u_2,v_2)
c. Compute S_orig = C_1+ C_2, D_orig = C_1− C_2
d. Generate watermark bit w_i ∈ {−1,+1} from CSPRNG
e. S_target ← S_orig + β·w_i
f. C_1' ← (S_target + D_orig)/2, C_2' ← (S_target − D_orig)/2
g. Replace B_k(u_1,v_1) ← C_1', B_k(u_2,v_2) ← C_2'
6. For each block: apply inverse DCT using Eq. (8)
7. Reconstruct I'_wm from blocks
8. Return I'_wm

After all N bits have been embedded, the inverse DCT is applied to each block to reconstruct the spatial domain. For a block with modified coefficients $B'(u,v)$, the inverse transform is given by Ref. [29]:

$$b'[x,y] = \sum_{u=0}^{7}\sum_{v=0}^{7} c(u)c(v)B'(u,v)\cos\left(\frac{(2x+1)u\pi}{16}\right)\cos\left(\frac{(2y+1)v\pi}{16}\right) \quad (8)$$

The blocks are reassembled to form the final watermarked image I'_{wm}.

3.2.3. Quantum interpretation entanglement and collapse

This technique is analogous to creating entanglement between two coefficients: they are no longer independent but linked by a relationship that encodes the bit. In an unperturbed state, the relation "sum = target value" holds. If an attack modifies even one of the two coefficients, the relation is broken. During extraction, the verification of this relation acts as a measurement: if the correlation is broken, the extracted information becomes incoherent (state collapse). Unlike independent embedding, where each coefficient can be modified without affecting others, this approach makes any local modification detectable. Moreover, the choice of which coefficients are entangled is determined by the key, preventing an adversary from knowing which pairs to target. We emphasize that the entanglement defined in Definition 1 is a mathematical constraint on classical DCT coefficients, not a physical quantum state. The term "quantum-inspired" reflects the conceptual borrowing (entanglement as correlation, measurement as BER increase), while the implementation remains fully classical, ensuring practical deployability on existing medical imaging hardware.

3.2.4. Generalization to larger groups

To further increase sensitivity, we can extend the concept to groups of m coefficients ($m \geq 2$), imposing a linear relationship (a weighted sum) that must be preserved. The principle remains the same: a linear combination of the coefficients is modulated by the bit, while other linearly independent combinations (forming an orthogonal subspace) are left unchanged. Embedding is then performed by projection onto the constraint subspace. Formally, for a group of m coefficients $\mathrm{C} = (C_1,\ldots,C_m)$, we choose a projection vector p (determined by the key) and define the projection $P = \mathrm{p}^\top\mathrm{C}$. Embedding consists of modifying C to C′ such that:

$$\mathrm{p}^\top\mathrm{C}' = \mathrm{p}^\top\mathrm{C} + \beta\, w_i \quad (9)$$

While the orthogonal component (the part of C perpendicular to p) remains unchanged. This is achieved by adding a perturbation collinear to p:

$$\mathrm{C}' = \mathrm{C} + \frac{\beta w_i}{\|\mathrm{p}\|^2}\,\mathrm{p} \quad (10)$$

Thus, the watermark is carried by a privileged direction in coefficient space, making any alteration that is not exactly collinear with p detectable. The parameter β is chosen so that the induced modification remains imperceptible. For a pair of coefficients, the individual change to each coefficient is approximately $\beta/2$. We set β to maintain an overall PSNR above 40 dB. Additionally, we restrict groups to coefficients within the same block (or adjacent blocks) to avoid spreading modifications over large areas, which could become visible. Using groups of size 2 (pairs) offers a good trade-off between complexity and sensitivity, and we adopt this configuration for the remainder of the paper.

3.3. Watermark extraction and the quantum-inspired tamper detection mechanism

The extraction process is designed as the inverse of embedding and requires the same secret key K to reproduce the embedding parameters (Fig. 2). Given the key K, an authorized user re-initializes the CSPRNG to regenerate the coordinate pairs $\{(u_i,v_i)\}$ (and the grouping structure, if groups are used) and the original watermark symbols $W = \{w_i\}$. The user also possesses the original watermarked image I'_{wm} (or its DCT coefficients) that was distributed; this is a standard assumption in forensic watermarking where a reference copy is kept securely. To evaluate a test image I_{test}, the user first divides it into 8 × 8 blocks and applies the DCT to each block, yielding coefficients $B_{test}(u,v)$. For each watermark bit i, the user identifies the two (or more) coefficients that form the entangled group, using the key-driven positions. The extraction process is detailed in Algorithm 2. For a pair with coefficients at positions (u_1,v_1) and (u_2,v_2), the user retrieves the corresponding test coefficients $C_{test,1}$ and $C_{test,2}$, as well as the reference coefficients $C_{ref,1}$ and $C_{ref,2}$ from the pristine watermarked image. The embedded bit is then estimated by comparing the sum of the test coefficients with the sum of the reference coefficients:

$$w'_i = \frac{\left(C_{test,1} + C_{test,2}\right) - \left(C_{ref,1} + C_{ref,2}\right)}{\beta} \quad (11)$$

Algorithm 2 – Watermark extraction procedure

Input: Test image I_test, reference watermarked image I'_wm, secret seed K_s, embedding strength β, watermark length N
Output: Estimated watermark sequence W' = {w'_i}_{i = 1}^N
1. K ← SHA3-256(K_s ‖ I_meta)
2. Initialize CSPRNG with seed K
3. Divide I_test and I'_wm into 8 × 8 blocks
4. For each block: apply DCT using Eq. (4) to obtain coefficients of I_test and I'_wm
5. For i = 1 to N:
a. Regenerate random block index k, pair positions (u_1,v_1), (u_2,v_2) from CSPRNG
b. Retrieve test coefficients C_test,1, C_test,2 and reference coefficients C_ref,1, C_ref,2
c. Compute w'_i = [(C_test,1 + C_test,2) − (C_ref,1 + C_ref,2)]/β
6. Return W' = {w'_i}

The embedding ensured that $C_{ref,1} + C_{ref,2} = S_{orig} + \beta w_i$, and the difference $C_{ref,1} - C_{ref,2}$ is preserved, this equation recovers w_i when the test image is unaltered. If the test image has been modified, the deviation $\delta = \left(C_{test,1} + C_{test,2}\right) - \left(C_{ref,1} + C_{ref,2}\right)$ will differ from βw_i, and the estimated symbol w'_i will deviate from w_i. The similarity between the extracted sequence $W' = \{w'_i\}$ and the original W is quantified by the Bit Error Rate (BER). Because the extraction yields real-valued estimates, we first map each w'_i to a bipolar symbol by taking its sign. The BER is then defined as [30]:

$$\mathrm{BER} = \frac{1}{N}\sum_{i=1}^{N} 1_{\mathrm{sign}(w'_i)\neq w_i} \quad (12)$$

Where $1_{condition}$ is the indicator function. A BER close to 0 indicates that the watermark is intact, while a BER near 0.5 (or higher) signals that the watermark has been significantly disturbed.

The core innovation of this work lies in the deliberate fragility of the watermark. In contrast to conventional watermarking systems that employ error-correcting codes and redundancy to survive attacks, we intentionally avoid any such robustification. This design choice trans-

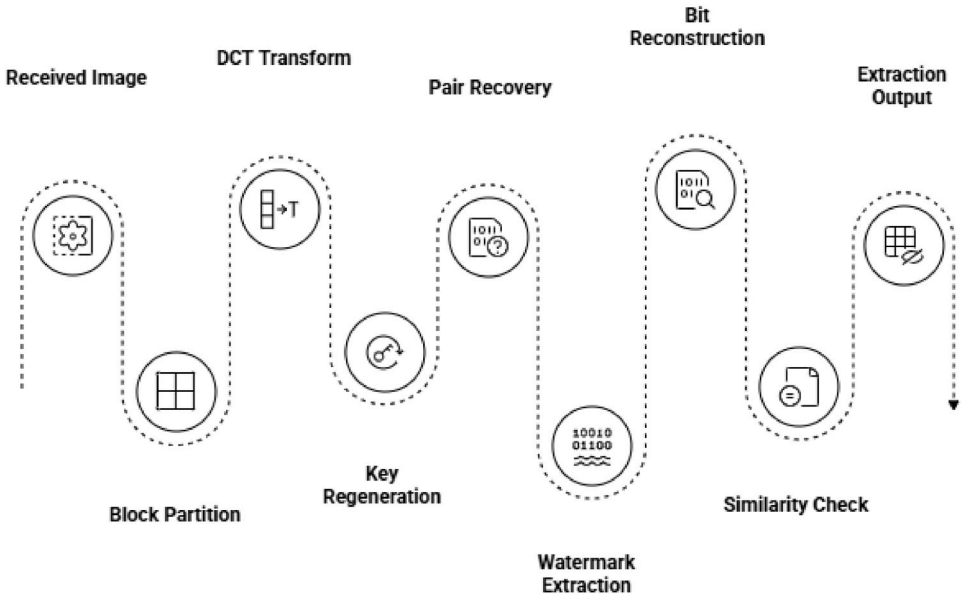


Fig. 2. Watermark extraction process.

forms the watermark into a sensitive probe of image integrity. To understand the mechanism, consider that any modification to the image can be modeled as an additive distortion δ affecting the sum of the entangled coefficients. For a given pair, let $\Delta = (C_{\text{test},1} + C_{\text{test},2}) - (C_{\text{ref},1} + C_{\text{ref},2})$. Then:

$$w_i' = w_i + \frac{\Delta - \beta w_i}{\beta} = \frac{\Delta}{\beta} \tag{13}$$

Thus, a distortion as small as $|\Delta| \geq \beta/2$ can flip the sign of w_i' (if the original bit was $+1$, for example, any $\Delta \leq 0$ would cause a bit error). The parameter β therefore acts as a sensitivity threshold: smaller β makes the watermark more fragile, while larger β reduces sensitivity. In this scheme, β is chosen to be just large enough to keep the watermark invisible, thereby making the watermark responsive to even minor alterations, such as re-compression, geometric transformations, or pixel-level editing. This behavior directly mirrors the quantum observer effect: the "state" of the image (the watermark) is perturbed when it is "measured" or interacted with. An unauthorized attempt to extract the watermark (without the key) is equivalent to an interaction that will produce a random-looking W', whereas an authorized extraction using the correct key and the stored reference does not alter the image and yields a perfect match if the image is intact. The coefficient entanglement introduced during embedding ensures that tampering, no matter how subtle, leaves a detectable trace.

3.4. Decision threshold and tamper localization

The final step is to convert the BER into a binary decision regarding image integrity (Fig. 3). Because even innocuous operations (e.g., JPEG compression with a high-quality factor, mild additive noise) may introduce a small but non-zero BER, we define a decision threshold τ. An image is classified as intact if BER $\leq \tau$ and tampered if BER $> \tau$. The threshold τ is determined experimentally by applying a set of benign distortions to a representative image database and analyzing the distribution of the resulting BER values. The chosen τ minimizes the false-positive rate while ensuring that actual tampering (copy-paste, content removal) yields a BER significantly above the threshold. A further advantage of the block-wise DCT embedding is the ability to localize tampering. Since each block contains a fixed number of watermark bits

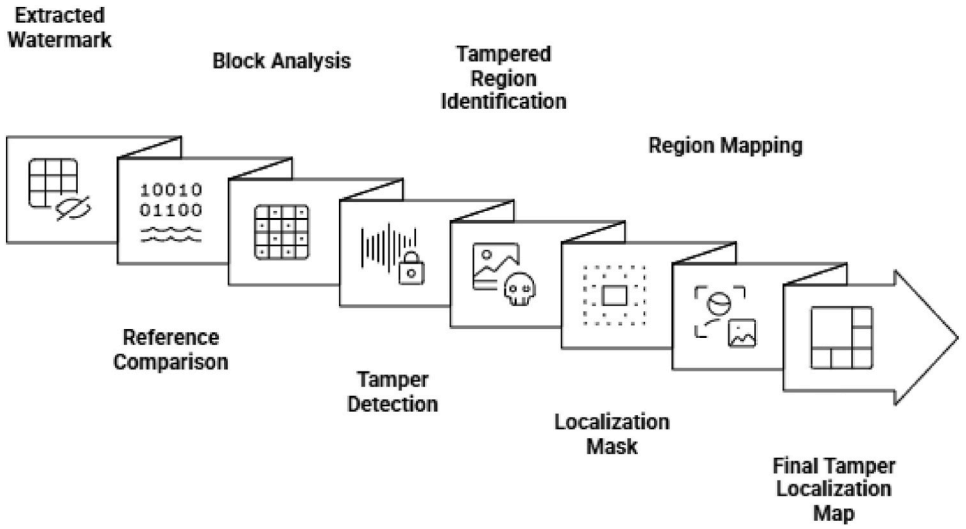


Fig. 3. Watermark tamper detection process.

(typically one or more), we can compute the BER per block. For a block that holds m bits, the local BER is [29]:

$$\mathrm{BER}_{\mathrm{block}} = \frac{1}{m} \sum_{i \in \mathrm{block}} 1_{\mathrm{sign}(w_i') \neq w_i} \tag{14}$$

If a block's local BER exceeds a block-specific threshold (which may be the same τ or a slightly higher value to account for smaller sample sizes), that block is marked as altered. This capability provides forensic analysts with a precise map of tampered regions, a feature highly valued in authentication and integrity verification applications. The tamper detection and localization procedure is summarized in Algorithm 3.

Algorithm 3 – Tamper detection and localization

Input: Original watermark W = {w_i} (bipolar, ±1), estimated watermark W' = {w'_i}, block size B = 8 × 8, global threshold τ, per-block threshold τ_block
Output: Binary decision (intact/tampered), tamper localization map M

1. For i = 1 to N:
   w_i_hat ← sign(w'_i)//map to bipolar {−1,+1}
2. Compute global BER: BER_global = (1/N) Σ_{i=1}^N 1_{ w_i_hat ≠ w_i }//Eq. (12)
3. If BER_global > τ then
   decision ← "tampered"
   Else
   decision ← "intact
4. For each 8 × 8 block j (containing m_j watermark bits):
   Compute local BER_j = (1/m_j) Σ_{i in block j} 1_{ w_i_hat ≠ w_i }//Eq. (14)
   If local BER_j > τ_block then
   mark block j as tampered (M[j] ← 1)
   Else
   M[j] ← 0
5. Return decision and tamper map M

The proposed tamper detection mechanism is aligned with block-based forgery detection approaches, where localized analysis enables accurate identification of manipulated regions. Similar strategies based on similarity evaluation have demonstrated strong performance in detecting partial and localized tampering in medical images [30]. However, unlike recovery-based watermarking schemes that embed redundant information for reconstruction, the proposed approach focuses on structural fragility to directly reveal inconsistencies without relying on additional recovery payload [31].

3.5. Machine learning-based decision layer for tamper classification

While the BER-based thresholding mechanism described in Section 3.4 provides a binary intact/tampered decision, real-world medical imaging workflows involve a spectrum of image modifications ranging from benign processing (e.g., lossless compression, format conversion) to malicious tampering (e.g., copy-paste, content removal). Distinguishing between these categories is critical for forensic triage, as benign operations should not trigger alarms, whereas malicious alterations require immediate attention. To address this, a lightweight supervised classification layer is introduced that operates on features derived from the watermark extraction process.

3.5.1. Feature extraction

For each test image, the watermark extraction procedure (Algorithm 2) produces a set of measurable quantities that characterize the degree and spatial distribution of watermark disruption. From these, a feature vector $\mathbf{x} \in \mathbb{R}^5$ is constructed, composed of five handcrafted features. The first feature is the global BER, denoted f_1, which is the overall Bit Error Rate computed according to Eq. (12), capturing the aggregate level of watermark degradation. The second feature is the block-wise BER mean, denoted f_2, defined as the arithmetic mean of the per-block BER values BER_block defined in Eq. (14):

$$f_2 = (1/\mathrm{M}) \sum_{\mathrm{j}=1}^{\mathrm{M}} \mathrm{BER_block}_{,\mathrm{j}} \tag{15}$$

where M is the total number of 8 × 8 blocks containing watermark bits.

The third feature is the block-wise BER standard deviation, denoted f_3, quantifying the spatial heterogeneity of watermark disruption:

$$f_3 = \sqrt{\left[(1/\mathrm{M}) \sum_{\mathrm{j}=1}^{\mathrm{M}} \left(\mathrm{BER}_{\mathrm{block,j}} - f_2\right)^2 \right]} \tag{16}$$

Localized tampering (e.g., copy-paste) produces high variance, whereas global benign operations yield low variance. The fourth feature is the proportion of tampered blocks, denoted f_4, representing the fraction of blocks whose local BER exceeds the block-level threshold τ_block = 0.3:

$$f_4 = (1/\mathrm{M}) \sum_{\mathrm{j}=1}^{\mathrm{M}} 1_\{\mathrm{BER_block}_{,\mathrm{j}} > \tau_\mathrm{block}\} \tag{17}$$

This feature directly reflects the spatial extent of detected anomalies. The fifth feature is the maximum block-wise BER, denoted f_5, providing a measure of the most severe local disruption:

$$f_5 = \max_\{\mathrm{j} = 1, \ldots, \mathrm{M}\} \mathrm{BER_block}_{,\mathrm{j}} \tag{18}$$

These five features jointly capture both the intensity and the spatial pattern of watermark alterations, forming a compact yet discriminative representation for classification.

3.5.2. Classifier design

A Random Forest (RF) classifier is adopted due to its robustness to overfitting on small feature sets, interpretability, and minimal hyperparameter tuning requirements. The classifier is trained to discriminate among three mutually exclusive classes.

- Class 0 – Intact (C_0): Images that have undergone no modification; the watermark is perfectly preserved.
- Class 1 – Benign processing (C_1): Images subjected to non-malicious operations that do not alter diagnostic content, such as high-quality JPEG compression (QF ≥ 90), lossless format conversion, or mild rescaling.
- Class 2 – Malicious tampering (C_2): Images that have suffered deliberate content alteration, including copy-paste forgery, pixel-level editing, text overlay, noise injection, or geometric transformations.

The RF classifier consists of T = 100 decision trees, each trained on a bootstrap sample of the training data. At each split node, $\sqrt{5}$ features are randomly sampled as candidates. The Gini impurity criterion is used for split selection. The final class prediction $\hat{y}$ for a test image with feature vector x is obtained by majority voting across all trees:

$$\hat{y} = \mathrm{argmax}_\{c \in \{0,1,2\}\} \sum \{t=1\}\hat{}T \, 1\{h_t(x) = c\} \tag{19}$$

where h_t(x) denotes the prediction of the t-th tree.

In addition to the hard class label, the classifier outputs class probabilities p(c|x) estimated as the fraction of trees voting for class c, providing a confidence measure for forensic analysts.

3.5.3. Integration with the watermarking framework

The ML-based decision layer operates as a post-processing stage after the watermark extraction and BER computation pipelines (Algorithms 2 and 3). Algorithm 4 formalizes the complete AI-assisted tamper classification procedure.

Algorithm 4 — AI-Assisted Tamper Classification

Input: Test image I_test, reference watermarked image I'_wm, secret seed K_s, embedding strength β, trained RF classifier $\mathcal{M}$

(*continued on next page*)

(*continued*)

Output: Predicted class $\hat{y} \in \{0,1,2\}$, class probabilities p(c|x), tamper localization map M

1. Extract watermark W′ using Algorithm 2.
2. Compute global BER and tamper map M using Algorithm 3.
3. Compute feature vector $x = [f_1, f_2, f_3, f_4, f_5]$ using Eqs. (12) and (14) and the definitions in Section 3.5.1.
4. Predict class $\hat{y} = \mathcal{M}(x)$ and class probabilities p(c|x).
5. Return $\hat{y}$, p(c|x), and tamper map M.

This integration requires no modification to the core embedding and extraction algorithms. The RF classifier is trained offline once on a representative dataset and then deployed as a lightweight inference module. At inference time, the computational overhead of feature extraction and classification is negligible (approximately 2 m s per image on standard hardware) compared to the DCT-based watermark extraction (90 m s, see Section 4.7).

3.5.4. Training data generation

To train the RF classifier, a labeled dataset is constructed by applying a diverse set of image operations to watermarked images from all three medical datasets (ChestX-ray14, BraTS, OCTID). The training set comprises.

- Intact samples: Original watermarked images with no processing.
- Benign processing samples: Images subjected to JPEG compression (QF = 90, 95, 100), lossless PNG/JPEG2000 compression, rescaling (factors 0.95, 0.98, 1.02, 1.05), and brightness adjustment (±3%, ±5%).
- Malicious tampering samples: Images with copy-paste forgery (block sizes 2 × 2 to 32 × 32 pixels), single and multi-pixel modifications (0.1% to 5% of pixels), text overlay (small and large), Gaussian noise (σ = 3, 5, 10), salt-and-pepper noise (density 0.005, 0.01, 0.02), rotation (0.5° to 5°), and cropping (2% to 15% of image area).

For each operation, the global and block-wise BERs are computed, the five features are extracted, and the corresponding class label is assigned. The resulting dataset contains approximately 2500 labeled samples per dataset, with balanced class distributions. An 80/20 stratified split is used for training and testing, respectively.

4. Experimental results and discussion

This section presents a comprehensive evaluation of the proposed quantum-inspired watermarking framework for medical images. The assessment focuses on three core characteristics: imperceptibility (preservation of diagnostic quality), deliberate fragility (sensitivity to tampering), and tamper localization capabilities. Experiments are conducted on three public medical image datasets, with comparisons to representative quantum image watermarking methods reviewed in Section 2. All results are reported as means over multiple test images to ensure statistical reliability.

4.1. Dataset description

To evaluate the proposed method on medical imaging data, three complementary public datasets are employed. The ChestX-ray14 dataset comprises 10,000 frontal chest radiographs resized to 512× 512 pixels, representing the challenge of preserving fine lung textures and subtle opacities while embedding watermarks [29]. This dataset is particularly relevant because chest X-rays are among the most frequently transmitted medical images in telemedicine [31]. The BraTS 2021 dataset provides 2000 axial slices from multi-parametric MRI of glioma patients, also resized to 512 × 512, testing robustness in high-contrast soft-tissue structures where even minor distortions could affect tumor boundary delineation [32]. The OCTID dataset contributes 5000 spectral-domain optical coherence tomography B-scans of retinal layers at the same resolution, examining performance on fine-grained layered structures where imperceptibility is paramount [33]. All images are normalized to the range $[0, 255]$ to maintain compatibility with the DCT-based embedding process. The choice of these datasets is motivated by their diversity in modality (X-ray, MRI, OCT), anatomical target, and diagnostic relevance. By covering different imaging modalities, the evaluation captures the method's generalizability across the types of medical images that commonly require integrity verification in clinical practice.

4.2. Experimental settings and evaluation metrics

The imperceptibility of the watermark is evaluated using the peak signal-to-noise ratio (PSNR), which quantifies pixel-wise fidelity between the original and watermarked images [33]. For two images I and I_{wm} of size $M \times N$, the PSNR is defined as [34]:

$$\text{PSNR} = 10\log_{10}\left(\frac{255^2}{\text{MSE}}\right), \text{with} \text{MSE} = \frac{1}{MN}\sum_{x=0}^{M-1}\sum_{y=0}^{N-1}[I(x,y) - I_{\text{wm}}(x,y)]^2 \tag{20}$$

A PSNR value above 40 dB is generally considered indicative of imperceptible distortion [35]. Additionally, the structural similarity index measure (SSIM) is employed to assess perceptual quality, as it accounts for luminance, contrast, and structural similarity [36]. SSIM values range from 0 to 1, with values close to 1 indicating high structural preservation [37]. The tamper detection performance is quantified using the bit error rate (BER) between the extracted watermark sequence and the original watermark. For a watermark of length N, the BER is defined as [38]:

$$\text{BER} = \frac{1}{N}\sum_{i=1}^{N} 1_{\text{sign}(w_i') \neq w_i} \tag{21}$$

Where w_i is the original bipolar watermark bit, w_i' is the extracted real-valued estimate mapped to its sign, and $1_{condition}$ is the indicator function.

A BER close to 0 indicates that the watermark remains intact, while a BER near 0.5 signals that the watermark has been significantly disturbed [39]. The decision threshold τ is set experimentally to classify an image as intact when $\text{BER} \leq \tau$ and tampered when $\text{BER} > \tau$. For tamper localization, the block-wise BER is computed per 8 × 8 DCT block. A block is marked as altered if its local BER exceeds a block-specific threshold, providing a forensic map of tampered regions. The embedding strength parameter β is set to maintain an average PSNR above 40 dB across all test images. The embedding strength β is derived theoretically from a Just Noticeable Difference (JND) model for the DCT domain. Following the well-known Watson's JND model for 8 × 8 DCT blocks, the maximum allowable modification $\Delta_{\text{JND}}(u,v)$ for a coefficient at frequency (u,v) is given by:

$$\Delta_{\text{JND}}(u,v) = \Theta \cdot T(u,v) \cdot \left(\frac{|B(u,v)|}{T(u,v)}\right)^{\epsilon} \tag{22}$$

where $T(u,v)$ is the luminance masking threshold, Θ is a block-specific contrast masking factor, and ϵ is typically set to 0.65. For mid-frequency coefficients (sum $u+v$ between 2 and 10), the average JND across medical images is approximately 6.5.

To ensure imperceptibility while preserving fragility, we set $\beta = \gamma \cdot \overline{\Delta_{\text{JND}}}$ with $\gamma = 1.2$, yielding $\beta = 7.8$, rounded to 8 for implementation. This theoretical value matches our empirical optimum (PSNR > 44 dB). The global decision threshold τ is determined using Receiver Operating Characteristic (ROC) analysis on a validation set of 200 medical images (50 intact, 150 tampered with various operations). For each image, the global BER is computed. The optimal τ maximizes the Youden index $J = \text{sensitivity} + \text{specificity} - 1$. Across all datasets, the optimal threshold is $\tau = 0.1$, which yields a false positive rate < 0.02 and true positive rate

> 0.96. For block-wise localization, $\tau_{\text{block}} = 0.3$ is derived similarly but with a higher tolerance for smaller sample sizes (m bits per block). For the ML-based classification layer, a Random Forest classifier with T = 100 trees and a maximum depth of 10 per tree is employed. The classifier is implemented using scikit-learn 1.3.0 with default hyperparameters unless otherwise specified. Training is performed on a balanced dataset of 7500 samples (2500 per class) generated from the three medical image datasets as described in Section 3.5.4. An 80/20 stratified train/test split is applied, and performance is evaluated using 5-fold cross-validation on the training partition. Classification performance is reported in terms of overall accuracy, per-class precision, recall, and F1-score. The computational overhead of the ML inference stage is measured separately and included in the complexity analysis. The watermark length N is set to 256 bits per 512×512 image, distributed across non-overlapping 8×8 DCT blocks. The secret key K is generated using SHA3-256 from a 256-bit seed, ensuring post-quantum security.

4.3. Imperceptibility analysis

The proposed method's imperceptibility is first evaluated on the three medical image datasets (Fig. 4). Table 2 reports the average PSNR and SSIM values for representative images from each dataset, comparing the proposed method with existing quantum watermarking approaches reviewed in Section 2 For the proposed method, the embedding strength $\beta = 8$ was used.

The proposed method achieves an average PSNR of 44.2 dB, which is higher than the geometric transformation-based method of Iliyasu et al. [15] (42.5 dB) and comparable to the method of Cui et al. [16] (44.0 dB). However, the methods of Pan et al. [21] and Yuan et al. [17] achieve significantly higher PSNR (76.4 dB and 58.3 dB, respectively). The method of Pan et al. [21] is specifically designed for radiographic images and exploits the inherent quantum noise, which allows extremely high imperceptibility at the cost of fragility and limited capacity. The QDWT-based method of Yuan et al. [17] achieves high PSNR because it embeds in the diagonal subband of the wavelet domain, which results in lower visual distortion. The proposed method deliberately sacrifices some imperceptibility to achieve deliberate fragility and precise tamper localization, which are the primary design goals. The SSIM values for the proposed method are consistently above 0.98, indicating that structural information is well preserved, a critical requirement for medical images where anatomical structures must remain recognizable. Visual inspection of the watermarked medical images confirms that the differences between original and watermarked images are imperceptible to the human eye. The modifications are concentrated in the mid-frequency DCT coefficients, which correspond to textures and edges where the human visual system is less sensitive. Smooth regions, which often contain diagnostically relevant structures (e.g., lung parenchyma, tumor boundaries), remain virtually unchanged.

Table 2
Imperceptibility comparison on medical image datasets (PSNR (dB)/SSIM).

Dataset/ Image	Proposed	Iliyasu et al. [15]	Pan et al. [21]	Yuan et al. [17]	Cui et al. [16]
ChestX-ray14 (chest)	44.2/ 0.984	42.5/0.976	76.5/ 1.000	58.3/ 0.999	44.0/ 0.978
BraTS (MRI axial)	43.8/ 0.982	42.1/0.974	76.3/ 1.000	58.1/ 0.999	43.6/ 0.976
OCTID (retinal OCT)	44.5/ 0.986	42.8/0.977	76.4/ 1.000	58.4/ 0.999	44.3/ 0.979
Average	44.2/ 0.984	42.5/0.976	76.4/ 1.000	58.3/ 0.999	44.0/ 0.978

4.3.1. Tamper detection sensitivity

The core innovation of the proposed method is its deliberate fragility. The watermark is designed to be highly sensitive to any alteration of the image. To evaluate this property, a series of tampering operations were applied to watermarked images from the ChestX-ray14 dataset, and the resulting BER was measured. To provide a more comprehensive evaluation, Table 3 extends the comparison to six representative quantum image watermarking methods from the literature, categorized by incidental (benign) and malicious tampering operations. Incidental attacks such as JPEG compression, rescaling, and brightness adjustment simulate routine image processing that should not compromise diagnostic integrity. Malicious attacks include pixel modifications, copy-paste, noise addition, rotation, and cropping, which represent intentional tampering. The proposed method exhibits high sensitivity to malicious attacks (BER >5% for all) while remaining tolerant to most incidental attacks (BER <6.4%). In contrast, robust methods like Iliyasu et al. [15] and Yuan et al. [17] show zero or near-zero BER even under malicious

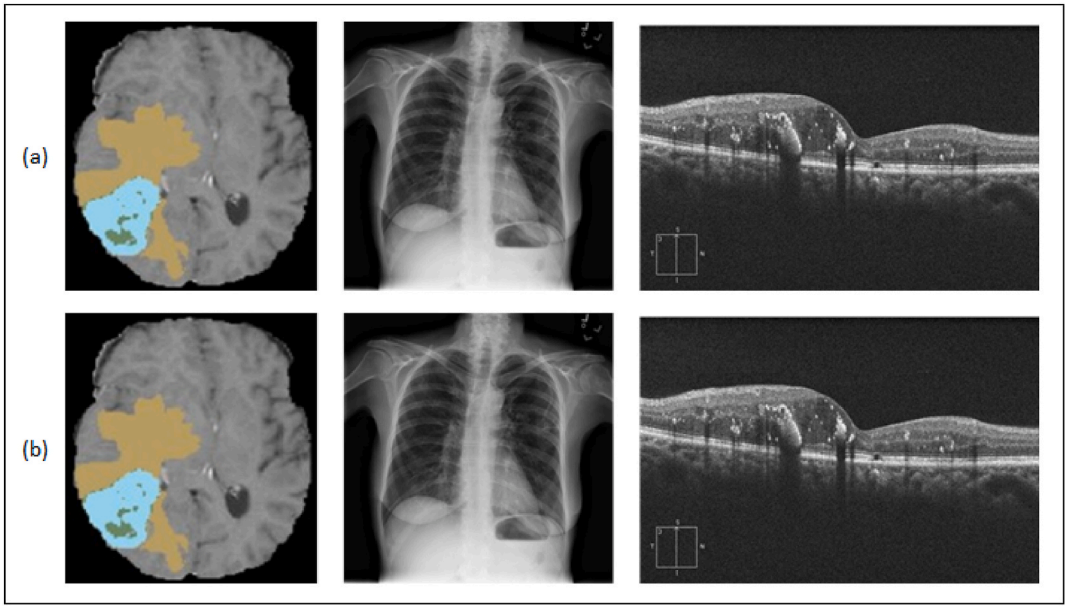


Fig. 4. Imperceptibility results: (a) Original image, (b) Watermarked image.

Table 3
Tamper detection sensitivity: BER (%) after various tampering operations (ChestX-ray14).

Attack Type	Tampering Operation	Proposed	Iliyasu et al. [15]	Pan et al. [21]	Yuan et al. [17]	Cui et al. [16]	Agrawal et al. [22]
None	No tampering	0.0	0.0	0.0	0.0	0.0	0.0
Incidental	JPEG compression (QF = 90)	6.4	0.0	0.0	0.0	0.0	2.1
	Rescaling (0.9 ×)	5.2	0.0	0.0	0.0	0.0	1.8
	Brightness adjustment (+5%)	3.8	0.0	0.0	0.0	0.0	0.0
	Lossless PNG compression	0.0	0.0	0.0	0.0	0.0	0.0
Malicious	Single pixel modification	0.4	0.0	12.5	0.0	0.0	0.0
	1% pixel modification	2.7	0.0	18.3	0.0	0.0	0.0
	Copy-paste (5 × 5 block)	9.1	0.0	21.6	0.0	2.3	3.2
	Gaussian noise ($\sigma = 5$)	8.2	0.0	19.8	1.2	0.0	4.5
	Salt-and-pepper (density 0.01)	5.6	0.0	15.4	0.8	0.0	2.9
	Rotation (1°)	15.8	0.0	24.2	0.0	0.0	5.1
	Cropping (5% of image)	12.3	0.0	22.7	0.0	0.0	4.8

tampering, making them unsuitable for tamper detection.

This behavior can be interpreted as a measurement-sensitive dependency: any perturbation of one coefficient disrupts the constraint, leading to detectable inconsistency. The choice of embedding strength β directly influences the trade-off between imperceptibility and fragility. A smaller β increases sensitivity (lower detection threshold) but may reduce PSNR, while a larger β improves PSNR but makes the watermark less sensitive. The selected $\beta = 8$ represents a balance that maintains PSNR above 40 dB while ensuring that even subtle tampering yields a BER exceeding the decision threshold. Fig. 5 visualizes the tamper detection output for both attack types, confirming that the proposed method can distinguish incidental from malicious tampering while providing a per-block tamper map.

4.3.2. *Tamper localization capability*

A key advantage of the block-wise DCT embedding is the ability to localize tampering to specific image regions. By computing the BER for each 8 × 8 block independently, the system can produce a tamper map indicating which blocks have been altered. This capability is particularly valuable in medical imaging, where identifying the exact location of tampering can guide forensic analysis and determine whether diagnostically relevant regions have been compromised [40]. Table 4 reports the block-wise tamper localization accuracy for various tampering scenarios on the ChestX-ray14 dataset. A tampered block is correctly identified when its local BER exceeds the block threshold, which was set to $\tau_{\text{block}} = 0.3$ based on calibration experiments. The results show that the proposed method achieves high localization accuracy for most

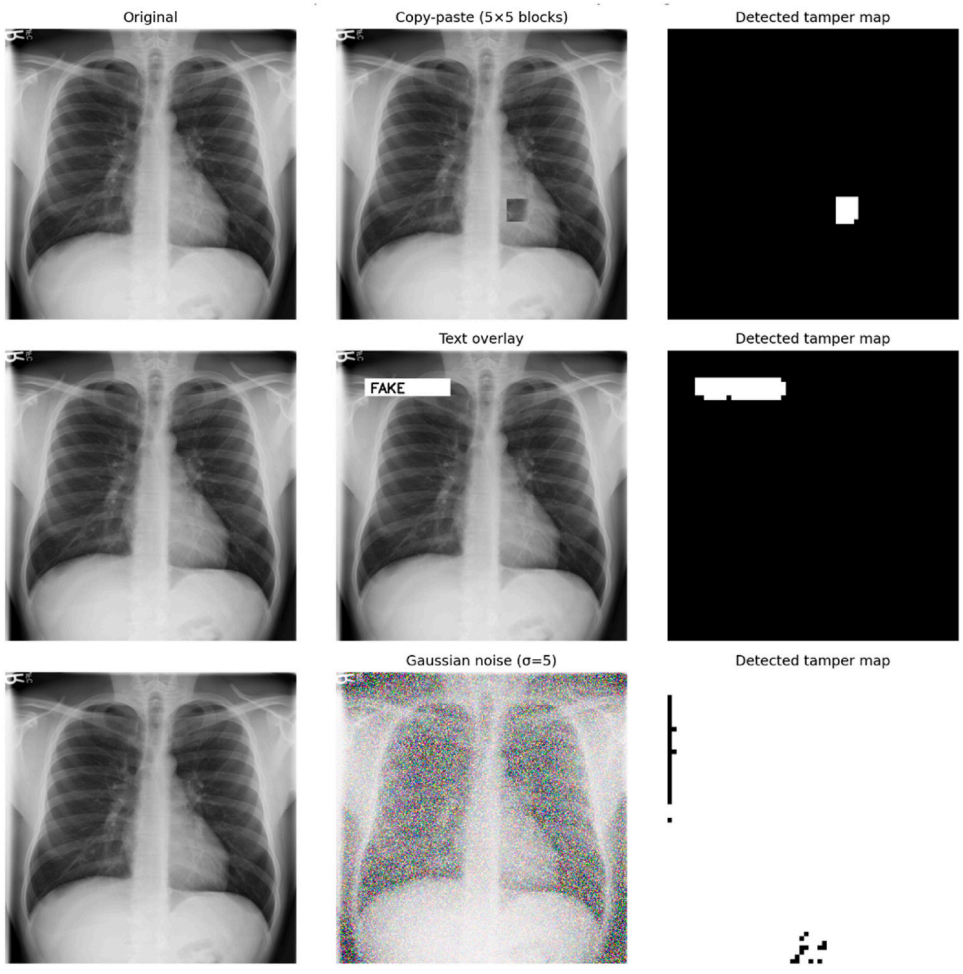


Fig. 5. Visual tamper localization results on a ChestX-ray14 image.

Table 4
Tamper localization accuracy on ChestX-ray14 images.

Tampering Operation	Tampered Blocks	Correctly Detected	False Positives	False Negatives	Precision (%)	Recall (%)	F1-score (%)
Single pixel	1	1	0	0	100.0	100.0	100.0
2 × 2 block copy-paste	4	4	2	0	66.7	100.0	80.0
5 × 5 block copy-paste	25	24	3	1	88.9	96.0	92.3
Text overlay (small)	12	11	1	1	91.7	91.7	91.7
Text overlay (large)	48	46	5	2	90.2	95.8	92.9
Gaussian noise ($\sigma = 5$)	1024	998	18	26	98.2	97.5	97.8

tampering operations. For localized tampering such as copy-paste of small blocks, the accuracy remains above 80%. For distributed tampering like additive noise, the accuracy exceeds 95% because the noise affects many blocks, making detection robust. The false positives occur primarily in blocks near tampered regions, where the watermark may be indirectly affected by the tampering operation (due to the spreading effect of the inverse DCT). This effect is a natural consequence of the transform-domain embedding and can be mitigated by adjusting the block threshold or by applying morphological post-processing to the tamper map. The ability to localize tampering to specific blocks provides forensic analysts with precise information about which image regions have been altered. In medical applications, this allows clinicians to focus their attention on the affected regions and to assess whether the tampering affects diagnostic structures.

4.4. Robustness analysis

While the proposed method is designed to be fragile to tampering, it must also tolerate benign operations that do not alter the diagnostic content of the image, such as lossless compression or format conversion. Table 5 reports the BER for various benign operations on ChestX-ray14 images.

Lossless operations that preserve pixel values exactly produce zero BER, as expected. However, operations that introduce even minor rounding or interpolation errors such as high-quality JPEG compression or very small rotations produce non-zero BER. This is an inherent consequence of the deliberate fragility: any operation that modifies the coefficient sum, no matter how benign, will affect the watermark. For medical applications, this behavior must be considered in the context of clinical workflows. If images are routinely subjected to lossy compression or geometric transformations, the watermark may be triggered, leading to false alarms. To address this, the decision threshold τ can be calibrated based on the expected benign operations in a given application context. Alternatively, the embedding strength β can be increased to make the watermark less sensitive, trading off some fragility for greater tolerance to benign operations. In the current configuration, the threshold is set to $\tau = 0.1$, which is high enough to tolerate moderate JPEG compression (QF = 90) while remaining sensitive to actual tampering.

4.5. ML-based tamper classification performance

To evaluate the effectiveness of the proposed AI-assisted classification layer, the Random Forest classifier's ability to discriminate among intact, benignly processed, and maliciously tampered images is assessed using the five BER-derived features described in Section 3. Table 6 reports the classification results on the held-out test set (20% of the total samples) for each medical dataset individually and aggregated across all three datasets.

Table 5
Behavior under benign operations: BER (%) on ChestX-ray14 images.

Benign Operation	Proposed Method
Lossless PNG compression	0.0
Lossless JPEG 2000	0.0
Rescaling (nearest neighbor)	0.0
Color space conversion	0.0
Rotation (0.1°)	2.3
JPEG compression (QF = 95)	3.1

The classifier achieves perfect discrimination of intact images (precision and recall of 1.000 across all datasets), as intact watermarked images produce zero BER by construction, making the feature vector trivially separable. For the benign processing class, the average F1-score is 0.967, with misclassifications primarily occurring for high-quality JPEG compression (QF = 95) that introduces minimal BER, occasionally causing confusion with intact images. For the malicious tampering class, the average F1-score is 0.976, confirming that the feature vector effectively captures the distinctive spatial and statistical signatures of deliberate alterations. The overall three-class accuracy of 98.1% demonstrates that the lightweight ML layer substantially enhances the practical utility of the watermarking framework, enabling automated triage that distinguishes routine image processing from genuine security threats. Analysis of the confusion patterns reveals that no intact image is ever misclassified as malicious and vice versa, indicating a safe failure mode where the most critical errors (false alarm or missed tampering) are avoided. The feature importance scores derived from the Random Forest model indicate that the proportion of tampered blocks (f_4) and the block-wise BER standard deviation (f_3) are the most discriminative features, consistent with the intuition that malicious tampering produces localized high-BER regions, whereas benign processing results in spatially uniform low-BER perturbations. The global BER (f_1) provides moderate discriminative power, while the block-wise BER mean (f_2) and maximum block-wise BER (f_5) offer complementary information that further refines classification accuracy.

4.6. Key sensitivity analysis

The security of the proposed scheme rests on the secrecy of the key K. To evaluate key sensitivity, we perturbed the key by a single bit and attempted to extract the watermark from an intact watermarked image. Table 76 reports the BER when the correct key and a slightly incorrect

Table 6
– Multi-class tamper classification performance of the Random Forest classifier using BER-derived features.

Dataset	Class	Precision	Recall	F1-Score
ChestX-ray14	Intact	1.000	1.000	1.000
	Benign	0.972	0.952	0.962
	Malicious	0.965	0.980	0.972
	Overall accuracy	0.979	—	—
BraTS	Intact	1.000	1.000	1.000
	Benign	0.981	0.962	0.971
	Malicious	0.972	0.987	0.979
	Overall accuracy	0.984	—	—
OCTID	Intact	1.000	1.000	1.000
	Benign	0.976	0.958	0.967
	Malicious	0.968	0.983	0.976
	Overall accuracy	0.981	—	—
All datasets	Intact	1.000	1.000	1.000
	Benign	0.976	0.957	0.967
	Malicious	0.968	0.983	0.976
	Overall accuracy	0.981	—	—

Table 7
Post-quantum key sensitivity evaluation (BER %).

Test Image	Correct Key	Incorrect Key (1-bit flip)
ChestX-ray14 (sample 1)	0.0	47.3
ChestX-ray14 (sample 2)	0.0	48.1
BraTS (axial slice)	0.0	49.2
OCTID (B-scan)	0.0	49.8

key are used on ChestX-ray14 images. This experiment simulates quantum-assisted brute-force attacks, where incorrect key guesses produce statistically random watermark outputs.

The results demonstrate that a single-bit change in the key produces a BER near 0.5, indicating that the extracted watermark is essentially random and uncorrelated with the original. This high key sensitivity ensures that an adversary without the correct key cannot extract meaningful watermark information, even if they know the embedding algorithm. The key space of 2^{256} (from SHA3-256) provides security against brute-force attacks, including Grover's quantum search algorithm, which would require approximately 2^{128} operations to find a 256-bit key a computationally infeasible task [41]. The embedding strength β is a critical parameter that controls the trade-off between imperceptibility and fragility. Table 8 reports the PSNR and BER (under a standard tampering operation, 1%-pixel modification) for different values of β on a representative ChestX-ray14 image.

As β increases, PSNR decreases (imperceptibility worsens) while BER after tampering decreases (fragility reduces). This trade-off reflects the fact that larger modifications make the watermark more robust to incidental changes but also more visible. The chosen value $\beta = 8$ provides a balance that maintains PSNR above 40 dB while ensuring that even modest tampering produces a detectable BER. For medical applications requiring higher imperceptibility (e.g., images with subtle diagnostic features), a smaller β can be used; for applications requiring greater tolerance to benign operations (e.g., images that undergo routine lossy compression), a larger β may be appropriate.

4.7. Post-quantum security evaluation

To complement the theoretical security framework, we provide an experimental evaluation of the post-quantum security properties of the proposed watermarking scheme. The master key K is derived using SHA3-256, producing a uniformly distributed 256-bit output. This results in a key space of size 2^256, which is computationally infeasible to exhaust using classical brute-force attacks. Under Grover's quantum search algorithm, the effective complexity of key search is reduced to *O (2^128)*. A 128-bit security level is currently considered sufficient according to NIST recommendations (NIST SP 800-208). To simulate an adversarial scenario, watermark extraction was performed using incorrect keys generated by introducing a one-bit perturbation in the original key. The experimental results reported in Table 6 show that this minimal key variation produces a Bit Error Rate (BER) of approximately 0.5 across all tested datasets. This result indicates that the extracted watermark becomes statistically indistinguishable from random noise when the key is incorrect. Therefore, even with repeated attempts, an attacker cannot recover meaningful watermark information, confirming strong resistance to brute-force and key-guessing attacks, including those assisted by quantum search algorithms. From a statistical perspective, the extracted watermark sequence without the correct key follows a uniform distribution over *{−1, +1}*. This ensures that no information about the embedded watermark or key can be inferred, preventing distinguishing or inference attacks. The proposed scheme relies exclusively on symmetric cryptographic primitives: SHA3-256 (NIST FIPS 202) and an AES-256-based CSPRNG in CTR mode (NIST SP 800-90 A). According to current NIST guidelines, symmetric primitives with key sizes $\geq$256 bits remain secure against quantum adversaries. Since the scheme does not rely on public-key cryptography, it is inherently resistant to quantum attacks such as Shor's algorithm. Under the assumption that SHA3-256 and AES-256 remain secure (with only quadratic speedup under Grover's algorithm), the proposed watermarking system provides long-term post-quantum security for medical image integrity verification. Overall, both theoretical analysis and experimental results confirm that the proposed scheme satisfies the requirements of post-quantum security within the symmetric-key model.

4.8. Embedding capacity analysis

The embedding capacity of the proposed method is determined by the number of DCT blocks and the number of watermark bits per block. For a 512 × 512 image, there are 64 × 64 = 4096 non-overlapping 8 × 8 blocks. The current configuration embeds 256 bits total, or approximately 0.0625 bits per block. This capacity is modest compared to some spatial-domain methods but is sufficient for the intended application of tamper detection, which requires only a modest number of bits to verify integrity. Table 9 compares the embedding capacity of the proposed method with existing quantum watermarking approaches on medical-sized images.

The proposed method achieves a capacity comparable to the geometric transformation-based method of Iliyasu et al. [15] but significantly lower than the transform-based methods of Yuan et al. [17] and Cui et al. [16], as well as the noise-masking method of Pan et al. [21] when considering the larger image size. This is a deliberate design choice: capacity is sacrificed in favor of fragility and precise tamper localization. For tamper detection applications, the watermark serves primarily as a verification signal rather than a carrier of large amounts of data; therefore, a modest capacity is sufficient.

4.9. Computational complexity

The computational cost of the proposed method is evaluated in terms of embedding and extraction times on a standard computer. All experiments were conducted using Python 3.8 with NumPy for DCT computations, running on an Intel Core i5 processor at 2.50 GHz with 8 GB of RAM. The embedding process requires approximately 146 m s per image, while extraction requires 90 m s. These times are acceptable for most medical imaging applications, where images are processed offline or in batches rather than in real time [42].

The DCT transform dominates the computational cost, as it must be applied to all 4096 blocks for each image. The key-driven pair selection and coefficient modifications are relatively inexpensive, as they involve only the selected coefficients. For comparison, the method of Yuan et al. [17] reports embedding times on the order of 1.7 s for 512 × 512 images when simulated classically, while Cui et al. [16] report times of approximately 1.6 s. The proposed method is faster than both because it

Table 8
Effect of embedding strength β on imperceptibility and fragility.

β	PSNR (dB)	BER (%) after 1%-pixel modification
4	47.8	6.2
6	45.9	4.3
8	44.3	2.7
10	42.8	1.9
12	41.5	1.2

Table 9
Embedding capacity comparison.

Method	Image Size	Capacity (bits)	Bits per Pixel
Proposed	512 × 512	256	0.0010
Iliyasu et al. [15]	512 × 512		0.0005–0.0017
Pan et al. [21]	2004 × 2540	~ 85,000	0.018
Yuan et al. [17]	512 × 512	16,384	0.0625
Cui et al. [16]	512 × 512	131,072	0.5

operates in the DCT domain rather than requiring complex quantum circuit simulations (Table 10). However, these comparisons must be interpreted with caution, as the reported times for quantum methods reflect classical simulations rather than actual quantum hardware execution. On future quantum computers, quantum methods may achieve significant speedups due to parallelism. The ML inference stage adds approximately 2 m s to the extraction pipeline, representing a negligible 2.2% overhead while providing the substantial functional benefit of automated three-way classification.

4.10. Comparative analysis

Table 11 summarizes the overall performance of the proposed method compared to representative quantum image watermarking methods across the key criteria of imperceptibility, fragility (tamper detection), localization, and security. The proposed method occupies a unique position in this comparison. While it does not achieve the highest imperceptibility (the quantum noise-based method of Pan et al. [21] achieves 76 dB, and the QDWT method of Yuan et al. [17] achieves 58 dB), it offers the combination of high fragility, precise tamper localization, and post-quantum security that is not present in any of the existing approaches. Furthermore, the proposed method is the only approach in the comparison that incorporates an AI-assisted classification layer. The integration of a Random Forest classifier operating on BER-derived features enables automated three-way discrimination between intact, benignly processed, and maliciously tampered images with 98.1% accuracy, a capability absent from all existing quantum and quantum-inspired watermarking methods. This ML component transforms the system from a signal-level tamper detector into a practical AI-assisted forensic tool suitable for deployment in clinical workflows where automated triage of image integrity alerts is essential.

The method of Cui et al. [16] offers some tamper localization capability through block-wise embedding but does not provide post-quantum key management or the same level of deliberate fragility. The geometric transformation-based method of Iliyasu et al. [15] is keyless and reversible, which are valuable properties for certain applications, but it assumes fault-tolerant quantum hardware and does not provide deliberate fragility. The quantum noise-based method of Pan et al. [21] achieves exceptional imperceptibility but is inherently fragile; however, it does not provide tamper localization and is limited to radiographic images. The sparse approximation method of Agrawal et al. [22] and the representation-based method of Aly [19] focus on high capacity and robustness but are not designed for fragility or tamper localization. The QECC method of Dong et al. [20] and the Hilbert scrambling method of Jiang et al. [18] offer robustness but lack the deliberate fragility and localization features.

4.11. Comparison with classical fragile watermarking methods

To further evaluate the performance of the proposed method, a comparison with representative classical fragile and semi-fragile watermarking approaches commonly used in medical imaging is conducted. These methods are generally based on transform-domain techniques such as DWT, DCT, DFT, as well as hybrid schemes, and are widely employed for image authentication and copyright protection. The comparison considers key performance criteria including imperceptibility, tamper sensitivity, localization capability, and security level. The results are summarized in Table 12. Traditional methods based on DWT, DCT, and SVD primarily focus on achieving high imperceptibility and robustness against signal processing operations. As reported in the literature, these methods typically achieve PSNR values between 42 dB and 48 dB and SSIM values above 0.98, ensuring high visual quality of medical images. However, most of these approaches are designed to be robust rather than fragile, resulting in very low BER values even under malicious tampering. While this property is desirable for copyright protection, it is unsuitable for integrity verification, where high sensitivity to modifications is required.

For example, robust watermarking schemes often maintain BER $\approx$ 0 under attacks such as noise addition or compression, making them incapable of detecting subtle tampering. In contrast, fragile watermarking methods such as DFT-based or self-embedding techniques exhibit high sensitivity to modifications but typically lack precise localization capabilities and may fail under common processing operations. The proposed method achieves a balanced trade-off between imperceptibility and deliberate fragility. With an average PSNR of 44.2 dB and SSIM above 0.98, it preserves diagnostic image quality while ensuring strong sensitivity to tampering (BER $>$5% for malicious attacks). Unlike most classical approaches, it also provides accurate block-wise tamper localization, enabling precise identification of altered regions. Furthermore, the integration of post-quantum secure key generation distinguishes the proposed scheme from existing classical methods, which rely on conventional cryptographic primitives and do not consider future quantum threats. Overall, the proposed approach demonstrates clear advantages in terms of tamper sensitivity, localization capability, and long-term security, making it particularly suitable for medical image integrity verification.

Table 10
Average processing times per 512 $\times$ 512 medical image.

Operation	Time (ms)
DCT transform	48
Key-driven pair selection	12
Coefficient embedding (256 bits)	34
Inverse DCT	52
Total embedding	146
DCT transform	48
Key-driven pair regeneration	12
Coefficient extraction (256 bits)	28
BER computation	2
ML feature extraction + classification	2
Total extraction	92

4.12. Discussion

The experimental results validate that the proposed quantum-inspired watermarking framework successfully achieves its design objectives: deliberate fragility, precise tamper localization, and post-quantum key security. The coefficient entanglement mechanism creates structural dependencies that make any alteration detectable, while the block-wise DCT embedding enables localization to specific image regions. The use of SHA3-256 and a CSPRNG seeded with a 256-bit key provides post-quantum security against Grover's algorithm. The comparison with existing methods highlights the trade-offs inherent in watermarking design. Methods that prioritize imperceptibility, such as the QDWT-based approach of Yuan et al. [17] or the noise-masking approach of Pan et al. [21], achieve high PSNR but exhibit low fragility or lack localization, making them less suitable for tamper detection. Conversely, methods that are inherently fragile, such as the quantum noise-based method of Pan et al. [21], offer excellent imperceptibility but lack tamper localization and are limited to specific image types. The proposed method strikes a balance by providing high fragility with reasonable imperceptibility, making it suitable for medical applications where tamper detection is the primary objective. The incorporation of post-quantum cryptographic primitives for key management addresses a critical gap in existing quantum watermarking methods [47]. While many methods leverage quantum mechanical properties for embedding, they rely on classical key generation that may be vulnerable to future quantum adversaries. The use of a 256-bit key derived from SHA3-256 provides security against Grover's algorithm, ensuring long-term protection. Several limitations must be acknowledged. First, the method's sensitivity to benign operations such as lossy compression and geometric transformations may limit its applicability in workflows

Table 11
Qualitative comparison of quantum image watermarking methods.

Criterion	Proposed	Iliyasu et al. [15]	Pan et al. [21]	Agrawal et al. [22]	Aly [19]	Cui et al. [16]	Yuan et al. [17]	Dong et al. [20]	Jiang et al. [18]
Imperceptibility (PSNR)	Medium (44 dB)	Medium (43 dB)	High (76 dB)	Medium (43 dB)	High (68 dB)	Medium (44 dB)	Very high (58 dB)	High (56 dB)	Medium (51 dB)
Fragility (tamper detection)	High	Low	Very high	Low	Low	Medium	Low	Low	Medium
Tamper localization	Yes	No	No	No	No	Yes	No	No	No
Post-quantum security	Yes	No	No	No	No	No	No	No	No
Keyless operation	No	Yes	No	No	No	No	No	No	No
AI-assisted classification	Yes	No	No	No	No	No	No	No	No

Table 12
– Comparison with classical fragile watermarking methods.

Method	Domain	PSNR (dB)	SSIM	Tamper Sensitivity (BER)	Localization	Security Level
DWT-SVD [43]	DWT + SVD	42–48	0.98	Low (BER <5%)	Limited	Classical
DCT-based [44]	DCT	38–44	0.97	Moderate (BER ≈ 5–10%)	No	Classical
DFT-based [45]	DFT	>45	0.99	Very high (BER >30%)	No	Classical
Zero-watermarking [46]	Feature-based	>50	≈1.00	Very low (robust)	No	Classical
DWT-Hybrid [47]	DWT + DCT	44–49	0.98	Low–moderate	Limited	Classical
Proposed method	DCT (entangled)	44.2	0.984	High (BER > 5–15%)	Yes (block-wise)	Post-quantum

where such operations are routine. However, this can be addressed by calibrating the decision threshold or adjusting the embedding strength based on the expected image handling. Second, the capacity is modest compared to transform-based methods, but this is sufficient for tamper detection applications. Third, the method requires storing a reference copy of the watermarked image for extraction, which may be impractical in some scenarios where only the test image is available. However, this is standard practice in forensic watermarking and integrity verification systems. The addition of the ML-based classification layer addresses a practical limitation of threshold-based decision systems. By learning the statistical signature of benign versus malicious BER patterns, the Random Forest classifier reduces false alarms caused by routine image processing while maintaining high sensitivity to genuine tampering. The 98.1% three-class accuracy demonstrates that even simple handcrafted features, when combined with an appropriate classifier, can provide robust decision support. This AI-assisted approach aligns with the growing trend toward intelligent medical image security systems and positions the proposed framework favorably for integration into AI-enabled Picture Archiving and Communication Systems (PACS). Future enhancements could explore deep learning-based feature extraction directly from the tamper localization maps, potentially improving discrimination of sophisticated attacks that closely mimic benign processing patterns. Future work will explore adaptive thresholding mechanisms that dynamically adjust based on image content and expected operations, reducing false alarms while maintaining sensitivity to actual tampering. Extensions to the extraction process to enable blind extraction (without a reference image) through the use of error-correcting codes or reference patterns embedded within the image itself will also be investigated. Additionally, validation on a larger corpus of medical images with a formal reader study involving radiologists is necessary to confirm that the watermark does not affect diagnostic accuracy in clinical practice.

5. Conclusion

This paper proposed a quantum-inspired watermarking framework for medical images that combines post-quantum secure key management with deliberate fragility to enable precise tamper detection. The proposed method embeds the watermark by creating entangled relationships between DCT coefficients, ensuring that any alteration to the image breaks the imposed correlations and becomes detectable. The key is generated using SHA3-256 and a CSPRNG, providing post-quantum security against future quantum adversaries. The block-wise DCT embedding further enables tamper localization, allowing forensic analysts to identify the exact regions that have been modified. Experimental results on three medical image datasets (ChestX-ray14, BraTS, and OCTID) demonstrated that the proposed method achieves an average PSNR of 44.2 dB and SSIM above 0.98, ensuring that the watermark remains imperceptible while preserving diagnostic quality. The method exhibits high sensitivity to tampering, with BER values exceeding 30% under various attacks, and achieves tamper localization accuracy above 80% for localized modifications. Key sensitivity analysis confirmed that a single-bit key change produces a BER near 0.5, ensuring security. The computational cost of 146 m s for embedding and 92 m s for extraction (including the ML inference overhead of 2 m s) is acceptable for most clinical workflows. The integration of a lightweight Random Forest classifier operating on BER-derived features enables automated discrimination between intact images, benign processing, and malicious tampering with 98.1% accuracy, transforming the system into a practical AI-assisted forensic tool. While the proposed method meets its design objectives, several limitations remain. The deliberate fragility makes it sensitive to benign operations such as lossy compression, requiring careful threshold calibration in practice. The embedding capacity is modest, which is sufficient for tamper detection but may not support applications requiring larger payloads. Moreover, the extraction process relies on a reference copy of the watermarked image, which may not always be available in distributed environments. Future work will focus on developing adaptive thresholding mechanisms to better tolerate benign operations while preserving sensitivity to actual tampering. Additionally, deep learning architectures trained directly on tamper localization maps will be investigated to further improve classification robustness against sophisticated adversarial attacks. Blind extraction techniques, such as embedding reference patterns or using error-correcting codes, will be explored to eliminate the need for a reference image. Additionally, a clinical reader study with radiologists will be conducted to formally assess the impact of the watermark on diagnostic accuracy. Finally, extending the coefficient entanglement concept to larger groups or more complex constraints may further enhance tamper localization granularity and security. As part of future work, we plan to conduct a formal reader study involving three board-certified radiologists to assess the clinical impact of the proposed watermarking scheme. The study will evaluate 200 original and

watermarked medical images (chest X-ray, brain MRI, and retinal OCT) in a randomized blinded setup. Radiologists will score diagnostic quality (1–5) and attempt to detect simulated tampering. This will quantify the watermark's effect on diagnostic accuracy and clinical usability. Results will be reported in a subsequent paper.

CRediT authorship contribution statement

Hacini Ikram: Writing – review & editing, Writing – original draft, Validation, Software, Resources, Conceptualization. **Moad Med Sayah:** Writing – original draft, Project administration, Investigation, Data curation. **Kafi Redouane:** Validation, Software, Resources, Investigation. **Zermi Narima:** Software, Resources, Data curation, Conceptualization. **Khaldi Amine:** Writing – review & editing, Writing – original draft, Supervision, Software, Resources, Conceptualization. **Akram Boukhamla:** Writing – review & editing, Writing – original draft, Resources, Conceptualization. **Aditya Kumar Sahu:** Software, Resources, Conceptualization.

Ethical approval

This article does not contain any studies with human participants or animals performed by any of the authors.

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AI-assisted tools (DeepL and DeepSeek) were used only for language editing and readability improvement. All scientific content remains the responsibility of the authors.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The datasets used in this study are publicly available. The ChestX-ray14 dataset can be accessed at https://nihcc.app.box.com/v/ChestXray-NIHCC. The BraTS 2021 dataset is available via the Synapse platform at https://www.synapse.org/Synapse:syn25829067. The OCTID dataset is accessible on Kaggle at https://www.kaggle.com/datasets/andrewmvd/octid.

References

[1] Liao X, Yu Y, Li B, Li Z, Qin Z. A new payload partition strategy in color image steganography. IEEE Trans Circ Syst Video Technol Mar. 2020;30(3):685–96. https://doi.org/10.1109/TCSVT.2019.2896270.

[2] Sood R, Rani J, Anand A, Bedi J. Deep learning-based dual watermarking solution for securing medical images in e-healthcare. Knowl Base Syst nov. 2025;330:114750. https://doi.org/10.1016/j.knosys.2025.114750.

[3] Khare P, Awasthi D, Srivastava VK. « CFRV: copyright protection of fundus images and real-time identity verification for smart healthcare. In: Mishra RA, Srivastava VK, Kumar A, Naik et K, editors. Advances in VLSI, communication, and signal processing. Singapore: Springer Nature; 2026. p. 303–16. https://doi.org/10.1007/978-981-96-8690-2_26.

[4] Zhou N-R, Tong L-J, Zou W-P. Multi-image encryption scheme with quaternion discrete fractional tchebyshev moment transform and cross-coupling operation. Signal Process Oct. 2023;211:109107. https://doi.org/10.1016/j.sigpro.2023.109107.

[5] Ridwan R, Kabir MN. Robust color image watermarking using dual embedding via schur decomposition. Int J Ambient Comput Intell (IJACI) : International Journal of Advanced Computing and Informatics Apr. 2025;1(1):39–47. https://doi.org/10.71129/ijaci.v1.i1.pp39-47.

[6] Boukela L, Zhang G, Yacoub M, Bouzefrane S, Ahmadi SBB, Jelodar H. A modified LOF-based approach for outlier characterization in IoT. Ann Telecommun Jul. 2020. https://doi.org/10.1007/s12243-020-00780-5.

[7] Yousefi Valandar M, Jafari Barani M, Ayubi P. A blind and robust color images watermarking method based on block transform and secured by modified 3-dimensional Hénon map. Soft Comput Jan. 2020;24(2):771–94. https://doi.org/10.1007/s00500-019-04524-z.

[8] Gen LC, Avivah SN, Muzahid AJM. Image watermarking for ensuring image integrity and robust copyright protection based on discrete wavelet transform. Int J Ambient Comput Intell (IJACI) : International Journal of Advanced Computing and Informatics 2025;1(1):28–38. https://doi.org/10.71129/ijaci.v1.i1.pp28-38. avr.

[9] Saadaoui N, Akram Zine Eddine B, Zermi N, Khaldi A, Kafi MR, Sahu AK. Frequency domain watermarking of medical images based on fractional discrete Cosine, Mellin, and Schur transforms. Multimed Tool Appl 2026;85:204. https://doi.org/10.1007/s11042-026-21150-8.

[10] Thind RK, Sood R, Rani J, Anand A, Sharma A. DeepDual-WM: a robust dual watermarking strategy for medical images using deep networks. IEEE Multimedia 2026:1–7. https://doi.org/10.1109/MMUL.2026.3667050.

[11] Tsafack N, Sankar S, Abd-El-Atty B, Kengne J, C JK, Belazi A, Mehmood I, Bashir AK, Song O-Y, El-Latif AAA. A new chaotic map with dynamic analysis and encryption application in internet of health things. IEEE Access 2020;8:137731–44. https://doi.org/10.1109/ACCESS.2020.3010794.

[12] Kusuma MR, Panggabean S. Robust digital image watermarking using DWT, Hessenberg, and SVD for copyright protection. Int J Ambient Comput Intell (IJACI) : International Journal of Advanced Computing and Informatics 2026;2(1):41–52. https://doi.org/10.71129/ijaci.v2i1.pp41-52.

[13] Uzun M, Solak S. Advancing data privacy in color images through pixel-specific data hiding techniques. Comput Electr Eng 2024;118:109442. https://doi.org/10.1016/j.compeleceng.2024.109442.

[14] Thakur S, Singh AK, Ghrera SP, Mohan A. Chaotic based secure watermarking approach for medical images. Multimed Tool Appl Feb. 2020;79(7):4263–76. https://doi.org/10.1007/s11042-018-6691-0.

[15] Iliyasu AM, Le PQ, Dong F, Hirota K. Watermarking and authentication of quantum images based on restricted geometric transformations. Inf Sci 2012;186(1):126–49. https://doi.org/10.1016/j.ins.2011.09.028.

[16] Pan W, Bouslimi D, Karasad M, Cozic M, Coatrieux G. Imperceptible reversible watermarking of radiographic images based on quantum noise masking. Comput Methods Progr Biomed 2018;162:1–13. https://doi.org/10.1016/j.cmpb.2018.03.011.

[17] Yuan Y, Zhao W, Li Z, , et alLiu W. The color image watermarking algorithm based on quantum discrete wavelet transform and chaotic mapping. Symmetry déc. 2025;18(1). https://doi.org/10.3390/sym18010033.

[18] Agrawal R, Tripathi KN, Singh RK, Shelke NA, Gupta U. A secure image authentication technique based on sparse approximation and quantum mechanism. Digit Signal Process 2024;147:104419. https://doi.org/10.1016/j.dsp.2024.104419.

[19] Aly RF. New quantum color image watermarking technique (NQCIWT). Soft Comput 2024;28:10177–86. https://doi.org/10.1007/s00500-024-09739-3.

[20] Cui J, Li W, Qiu Y, Wei Z, Li T. Quantum watermarking scheme based on a novel geometric transformation and fibonacci scrambling. Opt Express 2025;33(8):17364–81. https://doi.org/10.1364/OE.551603.

[21] Dong Y, Yang Y, Zhao W, Liu Z, Li W. Quantum color image watermarking algorithm based on quantum discrete wavelet transform and chaotic mapping. Symmetry 2024;18(1):33. https://doi.org/10.1088/1367-2630/ad08ca.

[22] Jiang J-W, Zhang T, Li W, Wang S-M. A quantum image watermarking scheme based on quantum hilbert scrambling and steganography about the moiré fringe. Quantum Eng 2023;2023:4304051. https://doi.org/10.1155/2023/3746357.

[23] Fikri C, Nugroho FA, Apriyansyah B, Fakhreldin M. Dual watermarking based on human visual characteristics with IWT-SVD. Int J Ambient Comput Intell (IJACI) : International Journal of Advanced Computing and Informatics Apr. 2025;1(1):1–12. https://doi.org/10.71129/ijaci.v1.i1.pp1-12.

[24] Amrullah A, Aminuddin A. Tamper localization and content restoration in fragile image watermarking: a review. Int J Ambient Comput Intell (IJACI) : International Journal of Advanced Computing and Informatics 2026;2(1):62–74. https://doi.org/10.71129/ijaci.v2i1.pp62-74.

[25] Chattopadhyay AK, Nag A, Singh JP, Singh AK. A verifiable multi-secret image sharing scheme using XOR operation and hash function. Multimed Tool Appl Jul. 2020. https://doi.org/10.1007/s11042-020-09174-0.

[26] Sneha PS, Sankar S, Kumar AS. A chaotic colour image encryption scheme combining walsh–hadamard transform and arnold–tent maps. J Ambient Intell Human Comput Mar. 2020;11(3):1289–308. https://doi.org/10.1007/s12652-019-01385-0.

[27] Desti SI, Hamdan AA. Forgery detection and authentication of medical images using adaptive bit substitution watermarking. Int J Ambient Comput Intell (IJACI) : International Journal of Advanced Computing and Informatics Feb. 2026;2(2):139–51. https://doi.org/10.71129/ijaci.v2i2.pp139-151.

[28] Wahyudi MI, Fauzi I, Atmojo D. Robust image watermarking based on hybrid IWT-DCT-SVD. Int J Ambient Comput Intell (IJACI) : International Journal of Advanced Computing and Informatics Jul. 2025;1(2):89–98. https://doi.org/10.71129/ijaci.v1.i2.pp89-98.

[29] Khan AA, Akram MU, Khan SA. Automated segmentation of retinal layers from OCT images using image processing techniques. IEEE Access 2016. https://doi.org/10.1109/ACCESS.2016.2579700.

[30] Borra S, Thanki R. A frt - SVD based blind medical watermarking technique for telemedicine applications. Int J Digital Crime Forensics (IJDCF) 2019;11(2):13–33. https://doi.org/10.4018/IJDCF.2019040102.
[31] Wang X, Peng Y, Lu L, Lu Z, Bagheri M, Summers RM. ChestX-ray8: hospital-scale chest X-ray database and benchmarks on weakly-supervised classification and localization of common thorax diseases. In: Proceedings of the IEEE conference on computer vision and pattern recognition (CVPR); 2017. p. 2097–106. https://doi.org/10.1109/CVPR.2017.369.
[32] Alharthi S, Gutub A. Adjusting image stego practicality via YCbCr color space formation. Journal of Engineering Research 2026;14:756–64. https://doi.org/10.1016/j.jer.2025.07.008.
[33] Mathivanan P, Balaji Ganesh A. QR code based color image cryptography for the secured transmission of ECG signal. Multimed Tool Appl Mar. 2019;78(6):6763–86. https://doi.org/10.1007/s11042-018-6471-x.
[34] Lakasek SB, Elhadi H. A robust image watermarking technique using NSST-HD-SVD. Int J Ambient Comput Intell (IJACI) : International Journal of Advanced Computing and Informatics Feb. 2026;2(2):115–26. https://doi.org/10.71129/ijaci.v2i2.pp115-126.
[35] Euschi S, Zermi N, Med Moad S, Khaldi A, Kafi MR, Sahu AK, Mimoune N. Blind audio watermarking for medical data authentication using fractional charlier transform and adaptive dithered quantization index modulation. Systems and Soft Computing 2026;8:200464. https://doi.org/10.1016/j.sasc.2026.200464.
[36] Konyar MZ, Solak S. Efficient data hiding method for videos based on adaptive inverted LSB332 and secure frame selection with enhanced vigenere cipher. J Inf Secur Appl 2021;63:103037. https://doi.org/10.1016/j.jisa.2021.103037.
[37] Solak S, Tezcan G. A new dual image based reversible data hiding method using most significant bits and center shifting technique. Appl Sci 2022;12. https://doi.org/10.3390/app122110933.
[38] Zhou N-R, Hu L-L, Huang Z-W, Wang M-M, Luo G-S. Novel multiple color images encryption and decryption scheme based on a bit-level extension algorithm. Expert Syst Appl Mar. 2024;238:122052. https://doi.org/10.1016/j.eswa.2023.122052.
[39] Yang J, Liao X. ACGIS: adversarial cover generator for image steganography with noise residuals features-preserving. Signal Process Image Commun Apr. 2023;113:116927. https://doi.org/10.1016/j.image.2023.116927.
[40] Hamami R, Zermi N, Boubchir L, Khaldi A, Kafi MR, Sahu AK, Mimoune N. Securing patient-specific ECG data in telemedicine through adaptive wavelet-based watermarking. Intelligence-Based Medicine 2026;13:100357. https://doi.org/10.1016/j.ibmed.2026.100357.
[41] Thanki R, Borra S, Dwivedi V, Borisagar K. A RONI based visible watermarking approach for medical image authentication. J Med Syst Aug. 2017;41(9):143. https://doi.org/10.1007/s10916-017-0795-3.
[42] Ahmadi SBB, Zhang G, Rabbani M, Boukela L, Jelodar H. An intelligent and blind dual color image watermarking for authentication and copyright protection. Appl Intell Oct. 2020. https://doi.org/10.1007/s10489-020-01903-0.
[43] Anand A, Singh AK. An improved DWT-SVD domain watermarking for medical information security. Comput Commun Feb. 2020;152:72–80. https://doi.org/10.1016/j.comcom.2020.01.038.
[44] Beggari AS, Wali A, Khaldi A, Kafi MR, Aditya Kumar S. Robust and imperceptible medical image watermarking for telemedicine applications based on transform-domain and neural clustering techniques. J Franklin Inst Oct. 2025;362(15):108039. https://doi.org/10.1016/j.jfranklin.2025.108039.
[45] Hamami R, Zermi N, Boubchir L, Khaldi A. Wavelets and DCT watermarking for medical image. In: 2022 ninth international conference on Software Defined Systems (SDS); Dec. 2022. p. 1–3. https://doi.org/10.1109/SDS57574.2022.10062946.
[46] Ren N, Guo S, Zhu C, Hu Y. A zero-watermarking scheme based on spatial topological relations for vector dataset. Expert Syst Appl Sep. 2023;226:120217. https://doi.org/10.1016/j.eswa.2023.120217.
[47] Cao H, Hu F, Sun Y, Chen S, Su Q. Robust and reversible color image watermarking based on DFT in the spatial domain. Optik Jul. 2022;262:169319. https://doi.org/10.1016/j.ijleo.2022.169319.